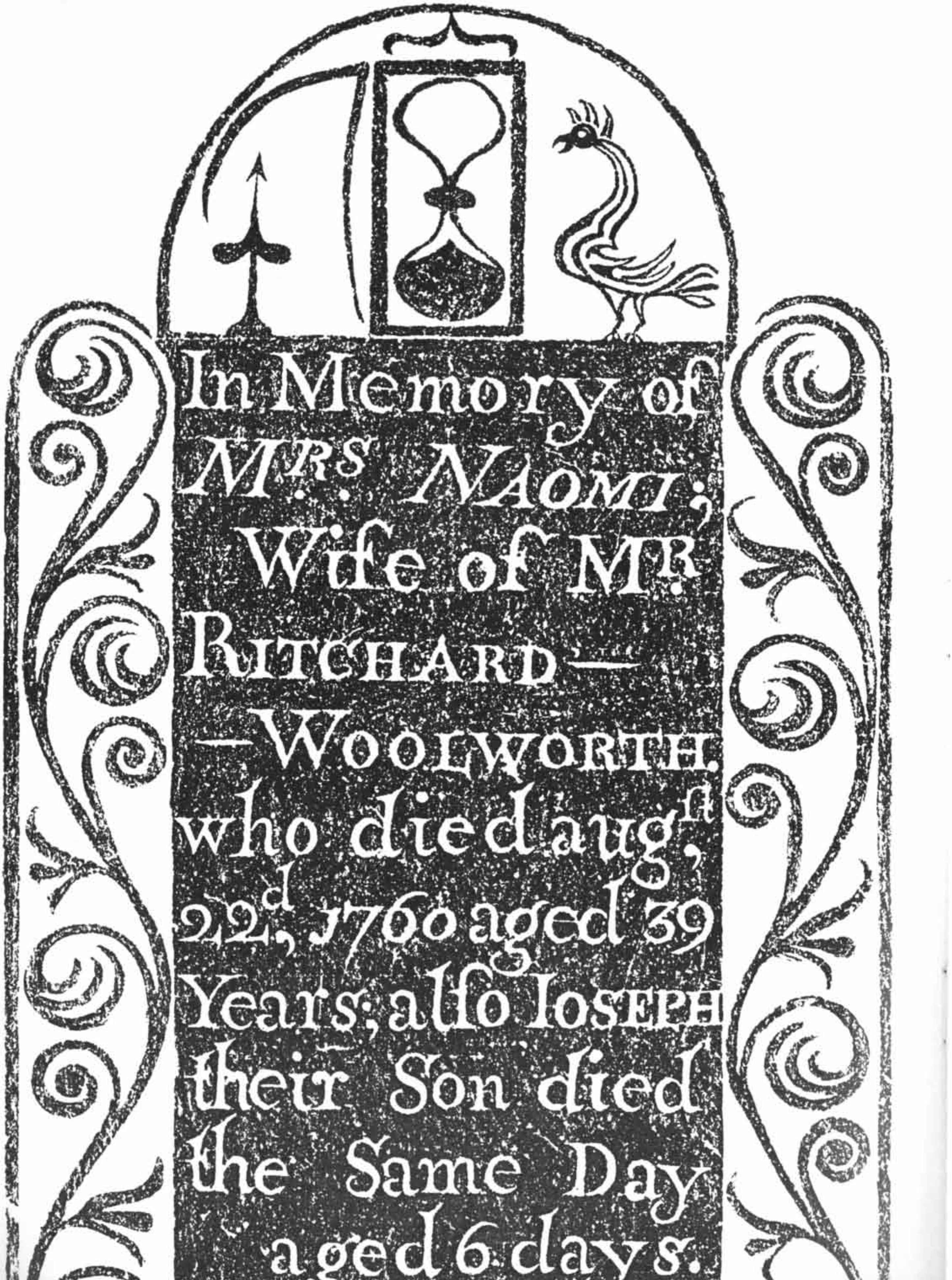

In Memory of
MRS. NAOMI;
Wife of MR.
RITCHARD—
—WOOLWORTH,
who died augst,
22d, 1760 aged 39
Years; alſo IOSEPH
their Son died
the Same Day
aged 6 days.

The History of the Human Population

Until some 200 years ago the size of the human population remained fairly stable because high birth rates were balanced by high death rates. The great demographic transition came when death rates fell

by Ansley J. Coale

In designating 1974 World Population Year the United Nations has given expression to worldwide interest in the rapid rate of population increase and to apprehension about the consequences of continued rapid growth. Much less attention is given to the growth of the population in the past, to the process by which a few thousand wanderers a million years ago became billions of residents of cities, towns and villages today. An understanding of this process is essential if one would evaluate the present circumstances and future prospects of the human population.

Any numerical description of the development of the human population cannot avoid conjecture, simply because there has never been a census of all the people in the world. Even today there are national populations that have not been enumerated, and where censuses have been taken they are not always reliable. Recent censuses of the U.S., for example, have undercounted the population by between 2 and 3 percent; some other censuses, such as the one taken in Nigeria in 1963, are evidently gross overcounts. Moreover, in many instances the extent of the error cannot be estimated with any precision.

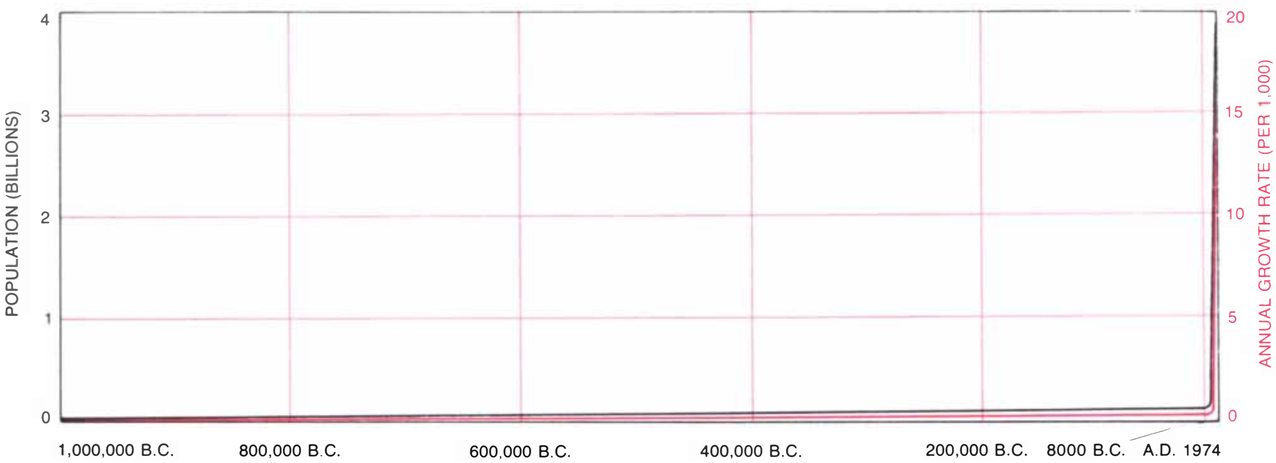
If the size of the population today is imperfectly known, that of the past is even more uncertain. The first series of censuses taken at regular intervals of no more than 10 years was begun by Sweden in 1750; the U.S. has made decennial enumerations since 1790, as have France and England since 1800. The census became common in the more developed countries only in the 19th century, and it has spread slowly to other parts of the world. India's population has been enumerated at decennial intervals since 1871, and a number of Latin American populations have been counted, mostly at irregular intervals, since late in the 19th century. The first comprehensive census of Russia was conducted in 1897, and only four more have been made since then. The population of most of tropical Africa remained uncounted until after World War II. A conspicuous source of uncertainty in the population of the world today is the poorly known size of the population of China, where the most recent enumeration was made in 1953 and was of untested accuracy.

As one considers earlier periods the margin of error increases. The earliest date for which the global population can be calculated with an uncertainty of only, say, 20 percent is the middle of the 18th century. The next-earliest time for which useful data are available is the beginning of the Christian era, when Rome collected information bearing on the number of people in various parts of the empire. At about the same time imperial records provide some data on the population of China, and historians have made a tenuous estimate of the population of India in that period. By employing this information and by making a crude allowance for the number of people in other regions one can estimate the population of the world at the time of Augustus within a factor of two.

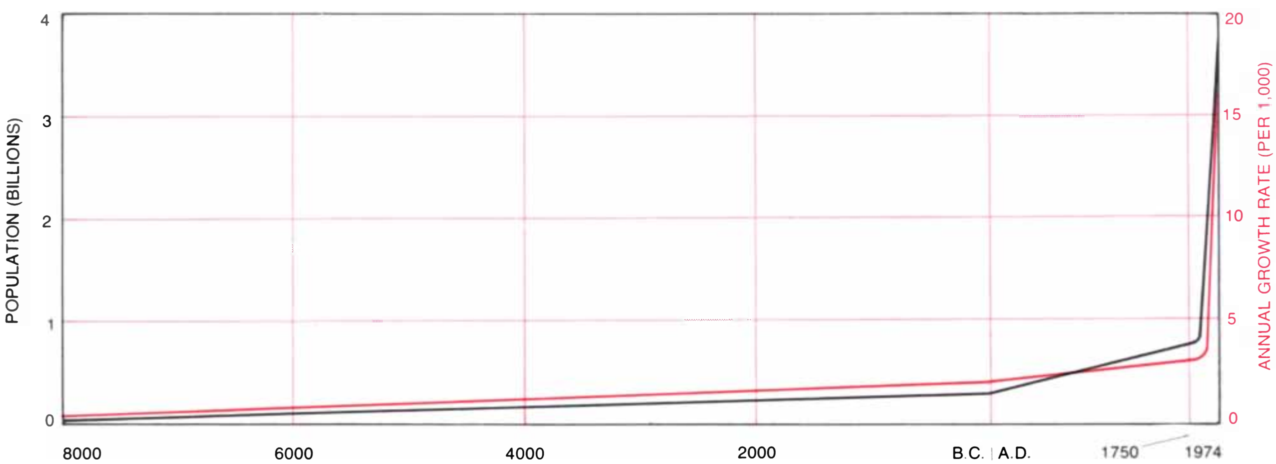
For still earlier periods the population must be estimated indirectly from calculations of the number of people who could subsist under the social and technological institutions presumed to prevail at the time. Anthropologists and historians have estimated, for example, that before the introduction of agriculture the world could have supported a hunting-and-gathering culture of between five and 10 million people.

From guesses such as these for the earlier periods and from somewhat more reliable data for more recent times a general outline of the growth of the human population can be constructed [*see illustrations on next page*]. Perhaps the most uncertain figure of all in these calculations is the size of the initial population, when man first appeared about a million years ago. As the human species gradually became distinct from its hominid predecessors there was presumably an original gene pool of some thousands or hundreds of thousands of individuals. The next date at which the population

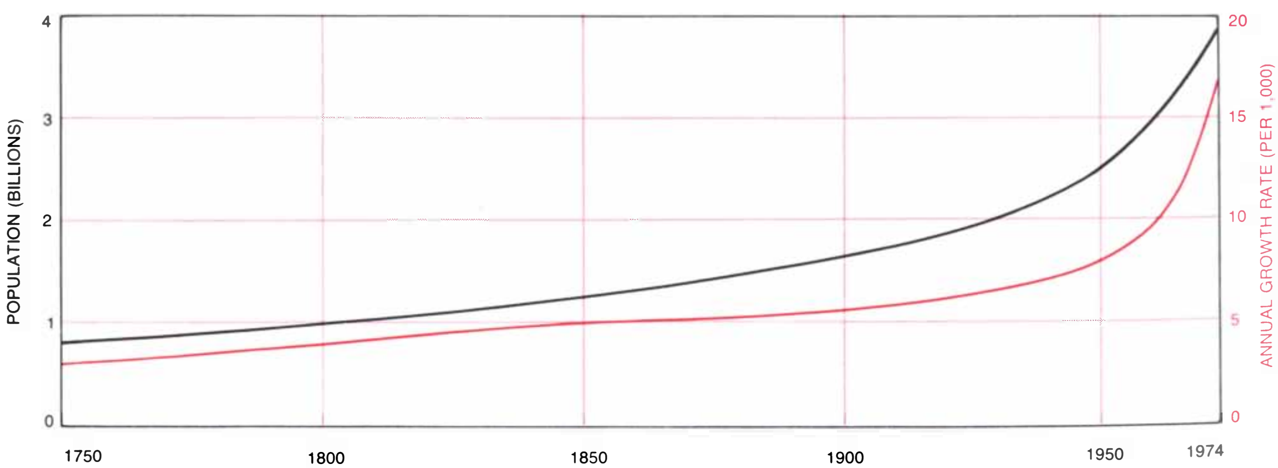
RUBBING OF A GRAVESTONE records the death of a mother and her child in 18th-century Massachusetts. The inscription reads (with emended punctuation and orthography): "In Memory of Mrs. Naomi, Wife of Mr. Ritchard, Woolworth, who died August 22d, 1760, aged 39 Years; also Joseph, their Son, died the Same Day aged 6 days." It is probable that both mother and son died as a result of some crisis attendant on childbirth, in the case of the mother perhaps from puerperal fever. Such deaths were very common throughout most of man's history; the high death rate they contributed to demanded that the birth rate also be high merely to sustain the population. A decline in the death rate, which had an important effect on the survival of infants and children, began in most parts of Europe and America in the decades following the events recorded on this gravestone. The figures at the top of the stone are a scythe and an hourglass, traditional symbols of mortality; a crowing cock, which probably represents an admonition to vigilance, and an object whose identity is uncertain but that may be a candle with snuffer, another commonplace figure in the imagery of death. The stone is at Longmeadow, Mass., and has been attributed to Aaron Bliss. The rubbing is reproduced from *Early New England Gravestone Rubbings*, by Edmund Vincent Gillon, Jr., published by Dover Publications, Inc. Surveys of gravestones are among the methods by which demographers measure historical populations.



OVERVIEW OF THE HUMAN POPULATION, from the emergence of man about a million years ago to the present, emphasizes the dichotomous nature of man's history. At this level of detail the growth curve approximates one where the size of the population (*black*) and the annual rate of increase (*color*) are constant for almost the entire period, then rise vertically in the most recent years.



INTRODUCTION OF AGRICULTURE some 10,000 years ago marks the beginning of a period that represents about 1 percent of that considered in the illustration at the top of the page. Even in this much briefer time span, however, the rate of population increase was modest throughout most of the period, and the gain during the past few centuries again appears to be almost vertical.



PERIOD SINCE 1750 is characterized by rapid and rapidly accelerating growth in the size of the world population. This period represents only about .02 percent of man's history, yet 80 percent of the increase in human numbers has occurred during it. Moreover, within this period the rate of increase has climbed most dramatically in very recent times: it has doubled in the past 25 years.

can be estimated is at the initiation of agriculture and the domestication of animals, which is generally believed to have begun about 8000 B.C. The median of several estimates of the ultimate size of the hunting-and-gathering cultures that preceded the introduction of agriculture is eight million. Thus whatever the size of the initial human population, the rate of growth during man's first 990,000 years (about 99 percent of his history) was exceedingly small. Even if one assumed that in the beginning the population was two—Adam and Eve—the annual rate of increase during this first long interval was only about 15 additional persons per million of population.

After the establishment of agriculture the growth of the population accelerated somewhat. The eight million of 8000 B.C. became by A.D. 1 about 300 million (the midpoint of a range of informed guesses of from 200 million to 400 million). This increase represents an annual growth rate of 360 per million, or, as it is usually expressed, .36 per 1,000.

From A.D. 1 to 1750 the population increased by about 500 million to some 800 million (the median of a range estimated by John D. Durand of the University of Pennsylvania). It was at this time that the extraordinary modern acceleration of population growth began. The average annual growth rate from A.D. 1 to 1750 was .56 per 1,000; from 1750 to 1800 it was 4.4 per 1,000, bringing the population at the end of this 50-year interval to about a billion. By 1850 there were 1.3 billion people in the world, and by 1900 there were 1.7 billion, yielding growth rates in the respective 50-year intervals of 5.2 and 5.4 per 1,000. (These totals too are based on estimates made by Durand.)

By 1950, according to the UN, the world population was 2.5 billion, indicating an annual growth rate during the first half of the 20th century of 7.9 per 1,000. From 1950 to 1974 the growth rate more than doubled, to 17.1 per 1,000, producing the present world population of 3.9 billion. The median value of several projections made by the UN in 1973 indicates that by 2000 the population will be 6.4 billion, an increase that implies an annual growth rate during the next 25 years of 19 per 1,000.

It is evident even from this brief description that the history of the population can be readily divided into two periods: a very long era of slow growth and a very brief period of rapid growth. An understanding of the development of the population during these two phases can be derived from a few simple mathematical relations involving the absolute size of the population, the growth rate and the factors that determine the growth rate.

Persistent growth at any proportionate rate produces ever increasing increments of growth, and the total, even at a relatively modest rate of increase, surpasses any designated finite limit in a surprisingly short time. An increasing population doubles in size during an interval equal to 693 divided by the annual rate of increase, expressed in additional persons per 1,000 population [*see illustration on page 46*]. Thus in the period from A.D. 1 to 1750, when the growth rate was .56 per 1,000, the population doubled about every 1,200 years; in the next few decades, when a growth rate of about 20 per 1,000 is anticipated, the population will double in 34.7 years.

The cumulative effect of a small number of doublings is a surprise to common sense. One well-known illustration of this phenomenon is the legend of the king who offered his daughter in marriage to anyone who could supply a grain of wheat for the first square of a chessboard, two grains for the second square and so on. To comply with this request for all 64 squares would require a mountain of grain many times larger than today's worldwide wheat production.

In accordance with the same law of geometric progression, the human population has reached its present size through comparatively few doublings. Even if we again assume that humanity began with a hypothetical Adam and Eve, the population has doubled only 31 times, or an average of about once every 30,000 years. This is another way of saying that the peopling of the world has been accomplished with a very low rate of increase, when that rate is averaged over the entire history of the species. The average annual rate is about .02 additional persons per 1,000. Even when only the more rapid growth of the past 2,000 years is considered, the average rate is modest. Since A.D. 1 the population has doubled no more than four times, or about once every 500 years, which implies an annual rate of 1.4 persons per 1,000.

In the context of these long-term averages the rate of growth today seems all the more extraordinary, yet the source of this exceptional proliferation is in the conventional mathematics of geometric series. The population of the world increases to the extent that births exceed deaths; the growth rate is the difference between the birth rate and the death rate. Another way of stating the relation is that the average rate of increase, over a long period, is dependent on the ratio of the sizes of successive generations. This ratio is approximately equal to the average number of daughters born to women who pass through the span of fertile years multiplied by the proportion of women surviving to the mean age of childbearing. This product specifies the average number of daughters born during the lifetime of a newborn female, after making allowance for those women whose biological fertility is abnormal and for those who die before reaching the age of childbearing. When the product is 1—signifying one daughter per woman, under the prevailing conditions of fertility and mortality—successive generations are the same average size. When the product is 2, the population doubles with each generation, or about every 28 years.

The fertility of a population can also be measured by the number of offspring, both sons and daughters, born per woman during a lifetime of childbearing; this number is called the total fertility rate. Mortality is summarized by the average age at death, or the average duration of life, which is expressed as the expectation of life at birth. In 1973 the total fertility rate of American women was 1.94; the expectation of life at birth was 75 years. Thus women experiencing 1973 birth rates at each age would bear an average of 1.94 children, and women experiencing 1973 death rates at each age would have an average duration of life of 75 years.

When the average life span is short, the proportion of women surviving to the mean age of reproduction is small. In fact, among populations for which there are adequate data there is a close relation between these two numbers, and we can with some confidence estimate the proportion of women surviving to become mothers from the average duration of life. Another predictable characteristic of the human population is the ratio of male births to female births; for any large sample it is always about 1.05 to 1.

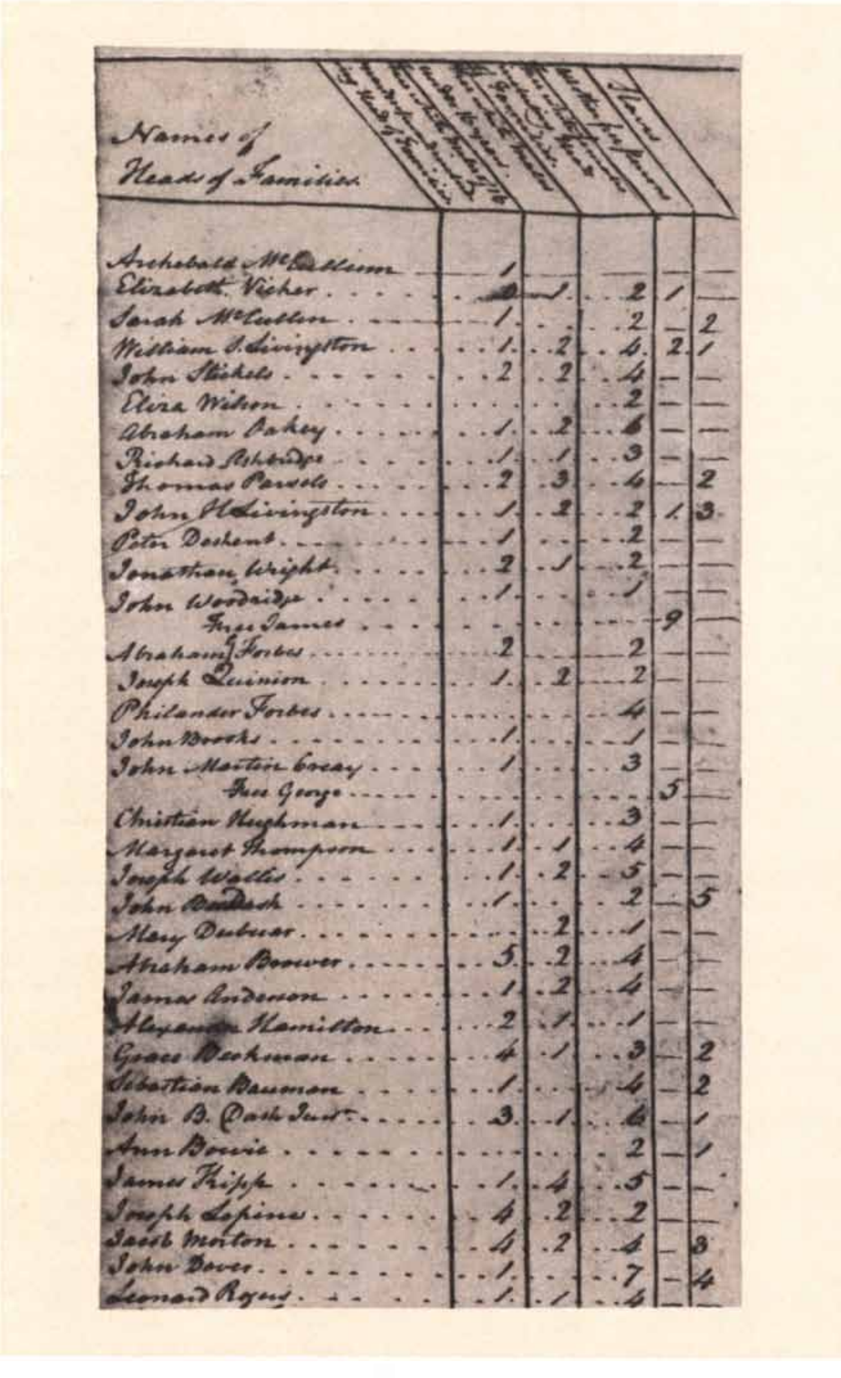
Because of these constant relations in the population it is possible to calculate all the combinations of female life expectancy and total fertility that will yield any specified growth rate. Of particular interest are the conditions producing zero population growth, since during most of the past million years the population has approached zero growth [*see illustration on page 45*]. In a static population the average duration of life is the reciprocal of the birth rate. Expressed another way, in a population of constant size the birth rate is the number of births per person-year lived and the average duration of life is the number of person-years lived per birth.

There are many combinations of fertility and mortality that will just maintain a population at fixed size. Consider a static population in which the average duration of female life is 70 years. Given this mortality rate, the proportion of women surviving to the mean age of childbearing is 93.8 percent. Because the size of the population is to remain constant the average number of daughters born per woman must be 1/.938, or 1.066; since there are 1.05 male births for each female birth, the total fertility rate must be 2.05 × 1.066, or 2.19. The birth rate in such a population is 1/70, or 14.3 per 1,000 population.

If the average duration of female life is 20 years, as it probably was at times during the premodern period, then 31.6 percent of the women survive to the mean age of childbearing and those who live to the age of menopause have an average of 6.5 children; the birth rate under these circumstances is 50 per 1,000. (It should be pointed out that there is no inconsistency in the survival of many women to menopause in a population in which the average age at death is 20 years. When the death rate is high, the average age at death is not at all a typical age at death. When the life expectancy in a static population is 20 years, for example, about half the deaths occur before age five, about a fourth occur after age 50, and only about 6.5 percent occur in the 10-year span centered on the mean age at death.)

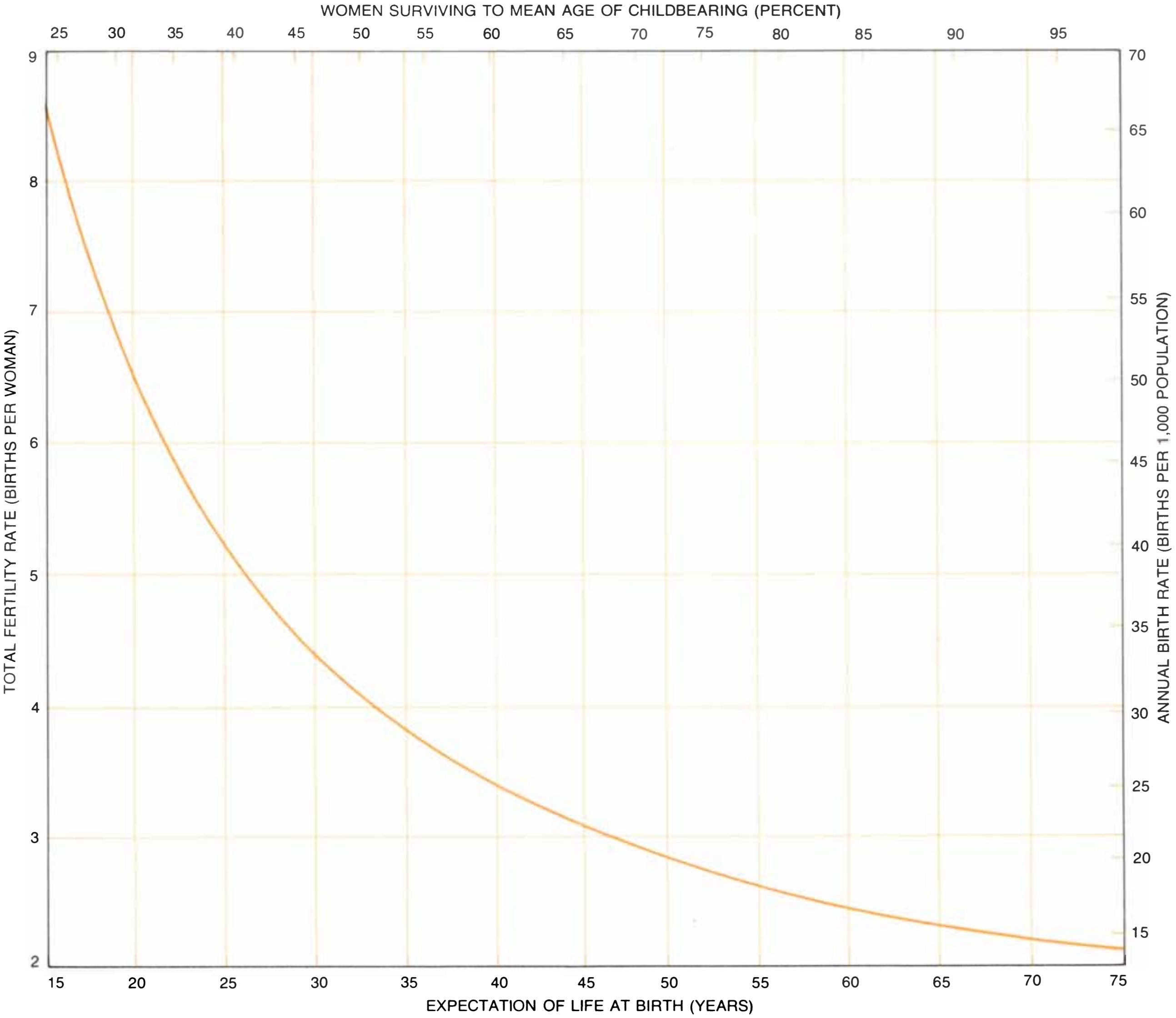
The importance of these relations is that they express the possible combinations of fertility and mortality that must have characterized the human population during each era of its history. If some other combination of fertility and mortality had been maintained for more than a few generations (as has happened during the past two centuries), the population would have expanded or contracted dramatically.

These combinations also determine the most extreme fertility and mortality rates possible in a static population. One limit is set by the minimum feasible mortality. When the average life expectancy is 75 years, 97.3 percent of all women survive to the mean age of reproduction, and it is necessary for them to have only 2.1 children to maintain the population; this represents a birth rate of 13.3 per 1,000. Any further reduction in mortality might raise the average duration of life to 80 years or more, but it would not significantly change the proportion of women surviving to childbearing age, nor would it much reduce the number of births per woman required to maintain the population. The other limit is imposed by fertility. When the life expectancy falls to 15 years, only 23.9 percent of all women live to have children, and those who do must have an average of 8.6 in order to prevent a decline in population. Although it is certainly biologically possible for a woman to bear more than eight or nine children, no sizable populations have been observed with



Names of Head of Families				All other free persons	Slaves
Archibald McCallum	1			—	—
Elizabeth Vicher	2	1	2	1	—
Sarah McCullen	1		2	—	2
William S. Livingston	1	2	4	2	1
John Stickels	2	2	4	—	—
Eliza Wilson			2	—	—
Abraham Oakey	1	2	6	—	—
Richard Ashbridge	1	1	3	—	—
Thomas Parsels	2	3	4	—	2
John H. Livingston	1	2	2	1	3
Peter Dochent	1		2	—	—
Jonathan Wright	2	1	2	—	—
John Woodridge	1		1	—	—
Free James				9	—
Abraham J. Forbes	2		2	—	—
Joseph Quinion	1	2	2	—	—
Philander Forbes			4	—	—
John Brooks	1		1	—	—
John Martin Creay	1		3	—	—
Free George				5	—
Christian Hughman	1		3	—	—
Margaret Thompson	1	1	4	—	—
Joseph Wallis	1	2	5	—	—
John [illegible]	1		2	—	5
Mary Dubear		2	1	—	—
Abraham Brower	5	2	4	—	—
James Anderson	1	2	4	—	—
Alexander Hamilton	2	1	1	—	—
Grace Beekman	4	1	3	—	2
Sebastian Bauman	1		4	—	2
John B. Cash Jun.	3	1	6	—	1
Ann Bowie			2	—	1
James Tripp	1	4	5	—	—
Joseph Lepine	4	2	2	—	—
Jacob Morton	4	2	4	—	3
John Dover	1		7	—	4
Leonard Rogers	1	1	4	—	—

CENSUS TALLY SHEET from the first enumeration of the U.S. population was employed in one of the earliest attempts to keep a current and regularly revised account of the size and distribution of a national population. Sweden, in 1750, was the first nation to institute a periodic census; the U.S. followed in 1790, when this schedule for a New York City neighborhood was filled out. The columns record the number of free white males 16 years old and older, free white males under 16 years old, free white females, other free persons and slaves. In the bottom half of the column appears the name of Alexander Hamilton.

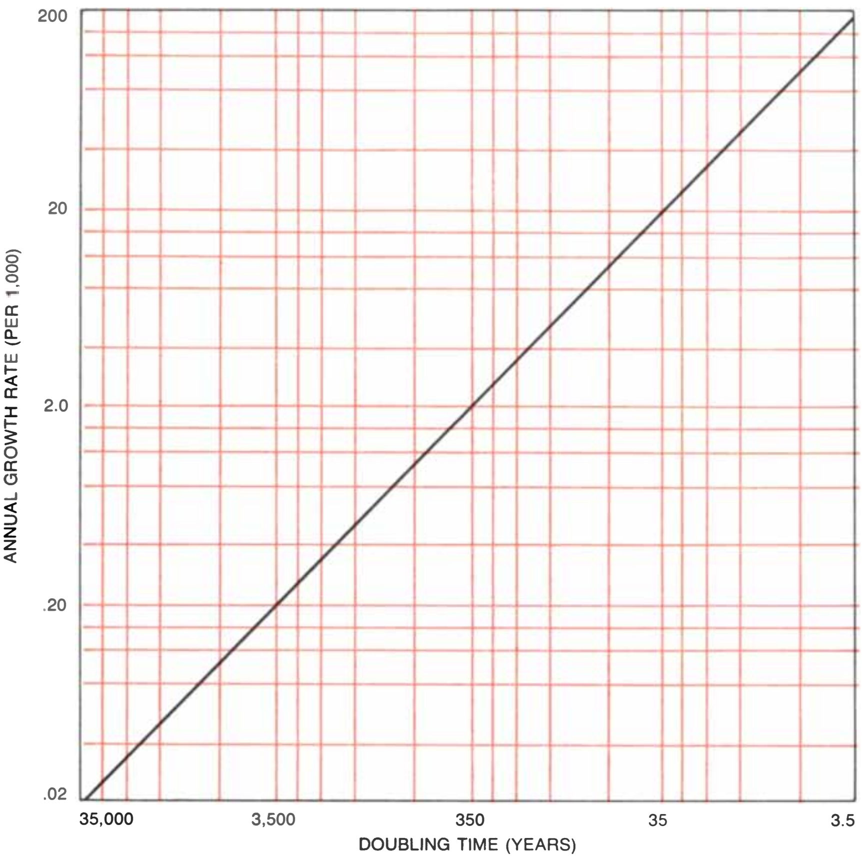


POPULATION IN EQUILIBRIUM can be maintained at constant size by many combinations of fertility and mortality. The total fertility rate is the number of children born per woman to a hypothetical group of women subject in each year of their lives to annual birth rates prevailing at a specified moment. Analogously, the expectation of life at birth is the average life span of a hypothetical group of people subject at each age to the death rates prevailing at a specified time. When the population is neither growing nor declining, two other demographic measures can be derived from these data: the birth rate and the proportion of women surviving to the mean age of childbearing. Some combination of these rates that approximates the conditions of zero growth must have prevailed during most of man's history. If the birth rate was 50 per 1,000, for example, then the average life span must have been 20 years, about a third of all women must have lived to childbearing age and those who survived must have had an average of 6.5 children.

total fertility much higher than eight births per woman.

Accurate records of human fertility and mortality are even more meager than records of numbers of people. Today fewer than half of the world population live in areas where vital statistics are reliably recorded; in most of Asia, almost all of Africa and much of Latin America, for example, the registration of births and deaths is inadequate. Precise information about fertility and mortality is therefore limited to the recent experience of the more developed countries, beginning in the 18th century in Scandinavia, the 19th century in most of the rest of Europe and the 20th century in Japan and the U.S. Much has been inferred about the present vital rates of underdeveloped countries from the age composition recorded in censuses, from the rate of population increase between censuses and from retrospective information collected in censuses and demographic surveys. For past populations, however, valid data on births and deaths are very rarely available, and they must therefore be derived by analyzing the forces that affect fertility and mortality.

Differences in fertility can be attributed to two factors: the differential exposure of women of childbearing age to the risk of childbirth through cohabitation with a sexual partner, and differences in the rate at which conceptions and live births occur among women who are cohabiting. In many populations the only socially sanctioned cohabitation is that between married couples, and thus the laws and customs governing the formation and dissolution of marriages influence fertility. A conspicuous example is the pattern of late marriage common until a generation ago in many Western European nations. For many years before World War II in Germany, Scandinavia, the Low Countries and Britain the average age of first marriage for women



DOUBLING TIME for a population is calculated by dividing the annual growth rate, in additional persons per 1,000 population, into the number 693. Until about 10,000 years ago the growth rate was .02 or less, and at least 35,000 years were required for the population to double. The rate today approaches 20 per 1,000, and the population could double in the next 35 years. Sustained growth has extreme consequences: 10 doublings, which would require 350 years at the present rate, would produce a population of more than four trillion.

was between 24 and 28, and from 1 to 30 percent remained unmarried at age 50. As a result the proportion of women of reproductive age who by being married were exposed to the risk of childbearing was less than half, and in some cases, such as Ireland, was as low as a third.

A much different nuptial custom that may also reduce fertility is common in areas of Asia and North Africa. Women are married at age 17 or 18, but the average age of the married male population is often eight or nine years greater than that of the married women. The fertility of some of the women is probably reduced by marriage to much older men, often widowers. Marriages are made by arrangement with the bride's parents, in many cases requiring the payment of a bride price, and older men are more likely to have the property or the prestige needed to claim the more desirable young women. Still another social influence on fertility is found in India, where Hinduism forbids the remarriage of widows. Although the prohibition has not always been scrupulously observed, it has doubtless reduced Indian fertility below what it might otherwise have been.

Among cohabiting couples fertility is obviously influenced by whether or not measures are employed to avoid having children. Louis Henry of the Institut National d'Études Démographiques has defined "natural fertility" as the fertility of couples who do not modify their behavior according to the number of children already born. Natural fertility thus defined is far from uniform: it is affected by custom, health and nutrition. Breast-feeding, for example, prolongs the period of postpartum amenorrhea and thereby postpones the resumption of ovulation following childbirth. In some populations low fertility can be attributed to pathological sterility associated with widespread gonorrheal infection. Finally, fertility may be influenced by diet, as has been suggested by the work of Rose E. Frisch and her colleagues at Harvard University. Age at menarche appears to be determined at least in part by the fat content of the body and is hence related to diet. Furthermore, among women past the age of menarche a sufficient reduction in weight relative to height causes amenorrhea. In populations with meager diets fertility may therefore be depressed. Because of the severe caloric drain of pregnancy and breast-feeding, it is probable that nursing prolongs amenorrhea more effectively in populations where the average body fat is near the threshold needed for a regular reproductive cycle.

The most conspicuous source of differences in fertility among cohabiting couples today is the deliberate control of fertility by contraception and induced abortion. In some modern societies very low fertility rates have been obtained: the total fertility rate has fallen as low as 1.5 (in Czechoslovakia in 1930, in Austria in 1937 and in West Germany in 1973).

The prevalence of birth-control practices is known from the direct evidence of fertility surveys for only a few populations, and for those only during the past two or three decades. (The International Statistical Institute has begun a World Fertility Survey that should illuminate present practices but not those of the past.) Indications that fertility was deliberately controlled in past societies must be inferred from such clues as the cessation of childbearing earlier among women who married early than among those who married late. Evidence of this kind, together with the observation of a large reduction in the fertility of all married women, indicates that birth control was common in the 17th century among such groups as the bourgeoisie of Geneva and the peers of France. Norman Himes, in his *Medical History of Contraception,* has shown that prescriptions for the avoidance of birth, ranging from magical and wholly ineffective procedures to quite practical techniques, have been known in many societies at least since classical Greek times. A doctoral dissertation at Harvard University by Basim Musallam has demonstrated that *coitus interruptus,* a contraceptive method that compares in effectiveness with the condom and the diaphragm, was common enough in the medieval Islamic world to be the subject of explicit provisions in seven prominent schools of law. On the other hand, analysis of parish registers in western Europe from the 17th and 18th centuries and observations in less developed countries today suggest that effective birth-control practices are not common in most rural, premodern societies.

Large fluctuations in fertility, and in mortality as well, are not inconsistent with the long period of near-zero growth that characterizes most of the history of the population. Although the arithmetic of growth leaves no room for a rate of increase very different from zero in the long run, short-term variations were probably frequent and of considerable extent. In actuality the population that from our perspective appears to have been almost static for hundreds of thousands of years may well have experienced brief periods of rapid growth, during which it expanded severalfold, and then suffered catastrophic setbacks. The preagricultural population, for example, must have been vulnerable to changes in climate, such as periods of glaciation, and to the disappearance of species of prey. Once the cultivation of crops had become established the population could have been periodically decimated by epidemics and by the destruction of crops through drought, disease or insect infestation. Moreover, at all times the population has been subject to reduction by man's own violence through individual depredation and organized warfare.

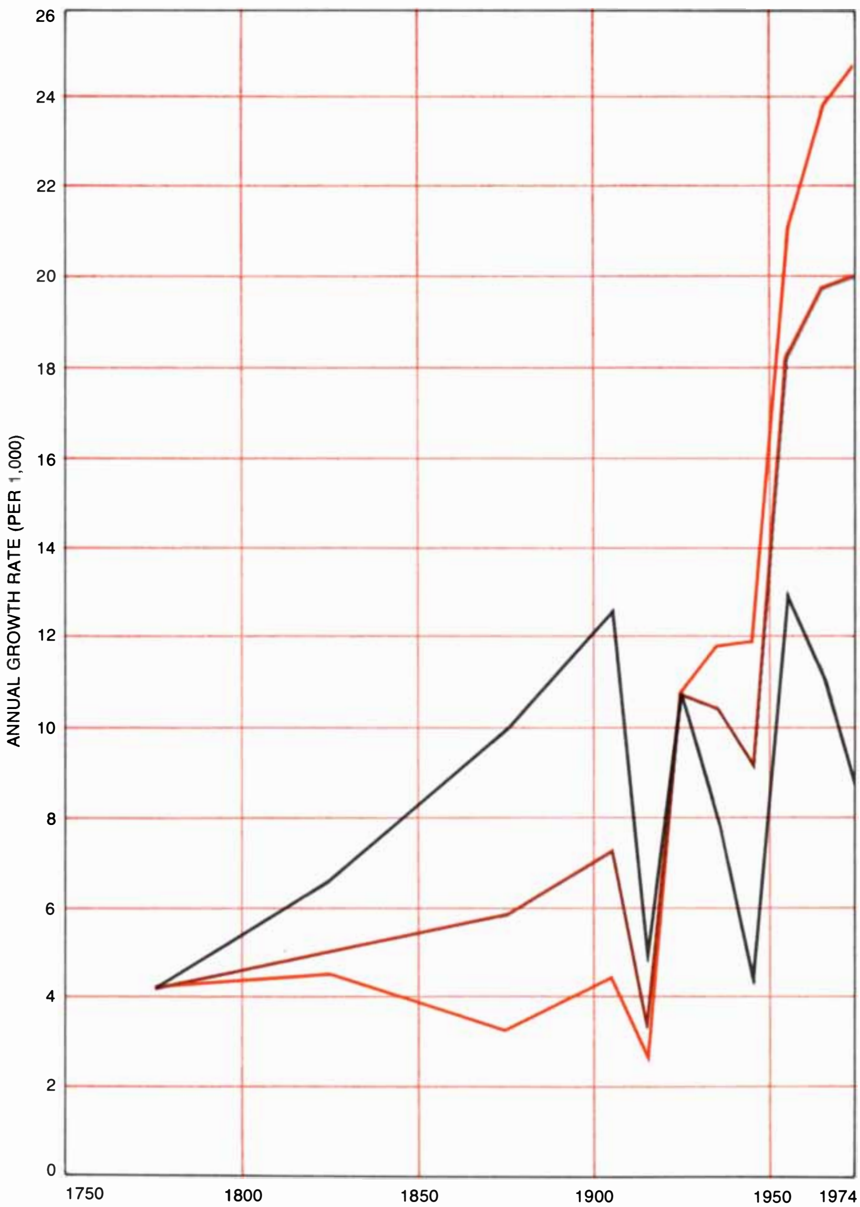
Because earlier populations never expanded to fill the world with numbers comparable to the billions of the 20th century, we must conclude that sustained high fertility was always accompanied by high average mortality. Similarly, sustained low fertility must have been compensated for by low mortality; any societies that persisted in low fertility while mortality remained high must have vanished.

In the conventional outline of human prehistory it is assumed that at each earlier date the average duration of life was shorter, on the principle that early man faced greater hazards than his descendants. It is commonly supposed, for example, that hunters and gatherers had higher mortality than settled agriculturists. The greater population attained by the agriculturists is correctly attributed to an enhanced supply of food, but the appealing inference that reduced mortality was responsible for this acceleration of growth is not necessarily justified.

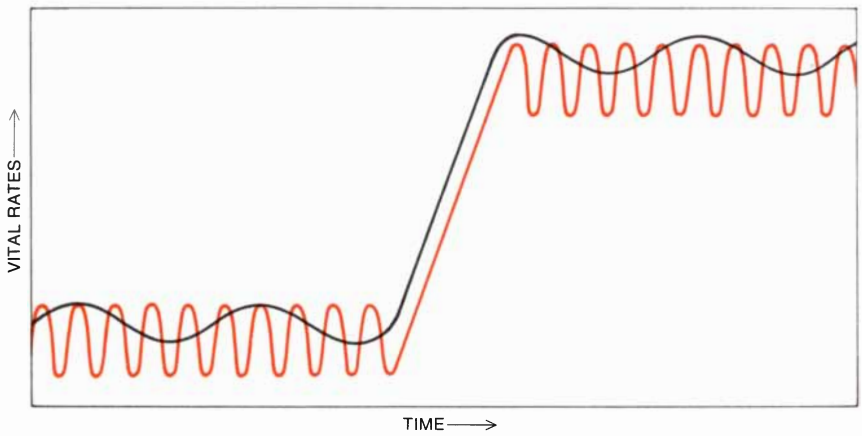
The advent of agriculture produced only a small increment in the growth rate; if this increment had been caused by a decline in mortality, the change in the average life expectancy would have been hardly noticeable. If in the hunting-and-gathering society the average number of births per woman was 6.5, for example, the average duration of life must have been 20 years. If the fertility of the early cultivators remained the same as that of their predecessors, then the increase in the life span required to produce the observed acceleration of growth is merely .2 year. The increase in life expectancy, from 20 to 20.2 years, would not have been perceptible.

If one assumes that preagricultural man had substantially higher mortality than the early cultivators, it must also be assumed that the hunters and gatherers had much higher fertility. If the earlier culture had an average age at death of 15 instead of 20, for example, then its fertility must have been 8.6 births per woman rather than 6.5. Such a change is not inconceivable; the complete reorganization of life represented by the adoption of agriculture could certainly be expected to influence both fertility and mortality. There is reason to suspect, however, that both vital rates increased rather than decreased [*see illustration on next page*].

Both disease and unpredictable famine might have increased the death rate of the first cultivators. Village life, by bringing comparatively large numbers into proximity, may have provided a basis for the transmission of pathogens and may have created reservoirs of endemic disease. Moreover, the greater density of agricultural populations may have led to greater contamination of food, soil and



DEVELOPED AND UNDERDEVELOPED NATIONS have different population histories. From the 18th century until after World War I the growth rate in the developed countries (*black*) exceeded that in the underdeveloped ones (*color*). Since the 1920's growth in underdeveloped regions has predominated, and since 1950 the gap has become large. Future trends (*black and color*) will be determined largely by events in the underdeveloped nations.



NEOLITHIC REVOLUTION had demographic consequences whose net effect was a slight increase in the rate of population growth. One reconstruction of these events suggests that the death rate (*black*) increased as a result of greater susceptibility to disease in village life, and perhaps also because agriculture is vulnerable to climatic crises. If the death rate did increase, then it is certain that the birth rate (*color*) also rose, and by a slightly greater margin. Both vital rates must have fluctuated from year to year, the death rate somewhat more than the birth rate. Even after the transition difference between the rates was small.

water. Greater density and the more or less total reliance on crops may also have made agriculturists extremely vulnerable to crop failure, whereas the hunting-and-gathering culture may have been more resistant to adversity.

If mortality did increase on the introduction of agriculture, then it is certain that fertility also rose, and by a slightly larger margin. The supposition that both vital rates did increase is supported by observations of the fertility rates of contemporary peoples who maintain themselves by hunting and gathering, such as the Kung tribe of the Kalahari Desert in southwestern Africa. Nancy Howell of the University of Toronto, analyzing observations made by her and by her colleague Richard Borshay Lee, has found that Kung women have long intervals between births and moderate overall fertility. A possible explanation, suggested by the work of Rose Frisch, is that the Kung diet yields a body composition low enough in fat to cause irregular ovulation. Interbirth intervals may be further prolonged by protracted breast-feeding combined with low body weight. If such conditions were common among preagricultural societies, the cultivation of crops could have increased fertility by increasing body weight and possibly by promoting the earlier weaning of infants so that mothers could work in the fields.

Unfortunately these speculations on the demographic events that may have accompanied the Neolithic revolution cannot be adequately tested by direct evidence. Until relatively recent times the only available indicators of mortality rates were tombstone inscriptions and the age-related characteristics of skeletons. Because the sample of deaths obtained in these ways may not be representative, it is not possible to reliably estimate for early periods such statistics as the average duration of life.

The accelerated growth in the world population that began in the 18th century is more readily understood if the areas classified by the UN as "more developed" and "less developed" are considered separately.

A general description, if not a full explanation, of the changing rates of increase in the more developed areas since the 18th century is provided by what demographers call the demographic transition. The changes in fertility and mortality that constitute the demographic transition are in general expected to accompany a nation's progression from a largely rural, agrarian and at least partly illiterate society to a primarily urban, industrial and literate one. Virtually all the populations classified by the UN as more developed have undergone demographic changes of this kind, although the timing and extent of the changes vary considerably.

The demographic experience common to all the more developed countries includes a major reduction in both fertility and mortality at some time during the past 200 years. In the 18th century the average duration of life was no more than 35 years, and in many of the nations that are now counted among the more developed it must have been much less. Today, almost without exception, the average life expectancy in these nations is 70 years or more. Two hundred years ago the number of births per woman ranged from more than 7.5 in some of the now more developed areas, such as the American colonies and probably Russia, to no more than 4.5 in Sweden and probably in England and Wales. In 1973 only Ireland among the more developed countries had a fertility rate that would produce more than three children per woman, and in most of the wealthier nations total fertility was below 2.5. Thus virtually all the more developed nations have, during the past two centuries, doubled the average life expectancy and halved the total fertility rate.

If the decline in fertility and mortality had been simultaneous, the growth in the population of the developed countries since 1750 might have been modest. Indeed, that was the experience of France, where the birth rate as well as the death rate began to decline before the end of the 18th century. As a consequence the increase in the French population was much less than that of most other European nations. The combined population of the developed countries experienced extraordinary growth after 1750, however, a growth that accelerated until early in the 20th century. The reason for the increase in numbers is that the decline in mortality has in almost all cases preceded the decline in fertility, often by many years [*see illustration on opposite page*].

The decline in fertility in the U.S., as in France, began early; it appears to have been under way by the beginning of the 19th century. Because of early marriage, however, fertility in the U.S. had been very high, so that the excess of births over deaths was still quite large. In most of the other more developed countries the birth rate did not begin to fall until late in the 19th century or early in the 20th.

Another universal feature of the transition is a change in the stability of the vital rates. In the premodern era the high birth rate was relatively constant, but the death rate fluctuated from year to year, reflecting the effects of epidemics and variations in the food supply. In those countries that have completed the demographic transition this pattern is reversed: the death rate remains constant but fertility varies considerably.

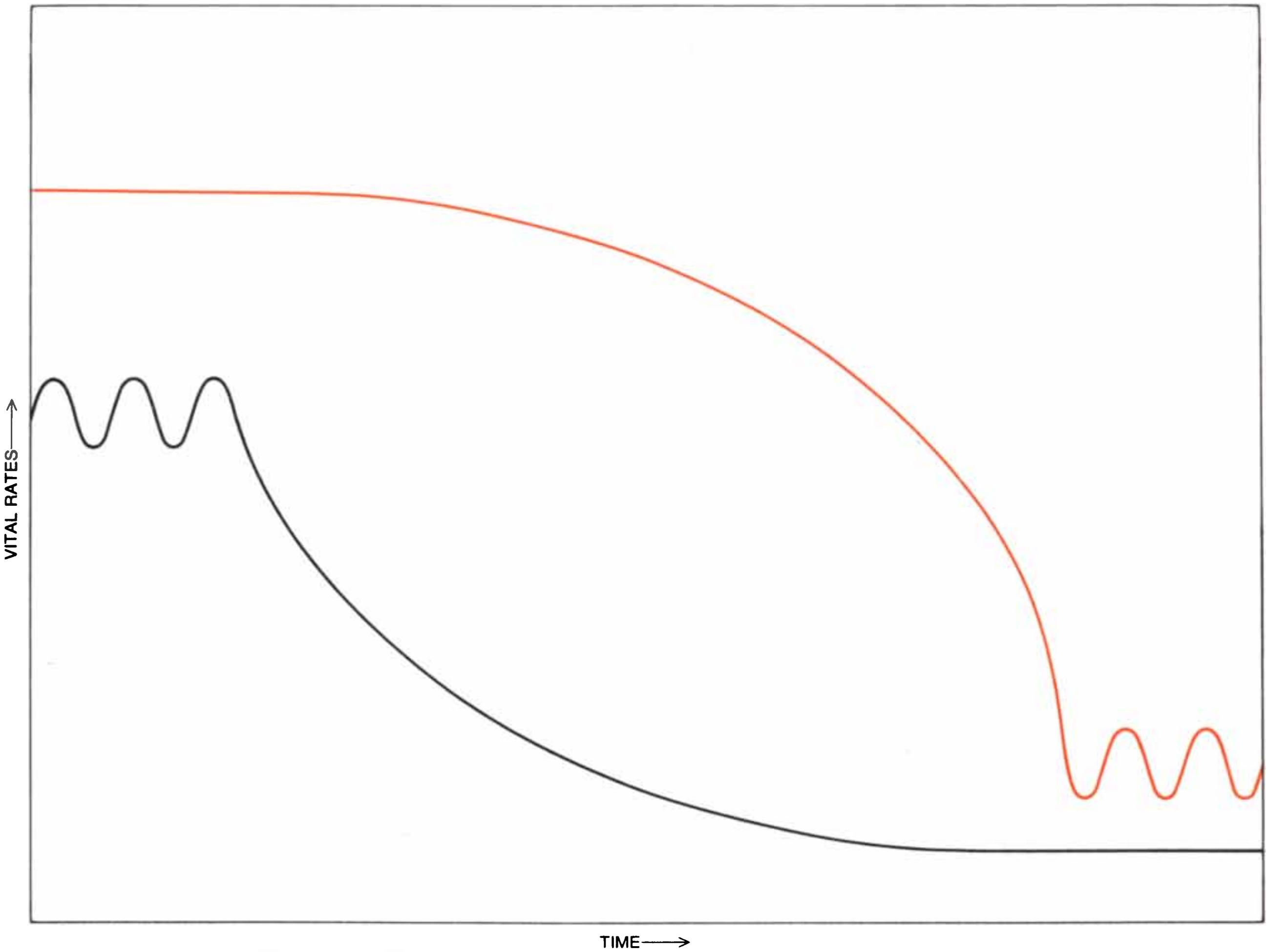
The causes of the event that began the demographic transition—the decline in mortality in the late 18th century—are a matter of controversy to social and medical historians. According to one school of thought, until the middle of the 19th century medical innovations in England could not account for the reduction of the English death rate; the principal factor proposed instead is an improvement in the average diet. Others argue that protection from smallpox through inoculation with cowpox serum, a procedure introduced late in the 18th century, was sufficient to markedly reduce the death rate. They propose that the further decline in mortality in the early 19th century may have been brought about by improvements in personal hygiene.

A third hypothesis is that before the 18th century fortuitous periods of low mortality were not exceptional, but that they were followed by periods of very severe mortality caused by major epidemics. According to this view, the late 18th century was a normal period of respite, and improved conditions early in the 19th century averted the next cycle of epidemics, which would otherwise have produced a recurrence of high mortality rates.

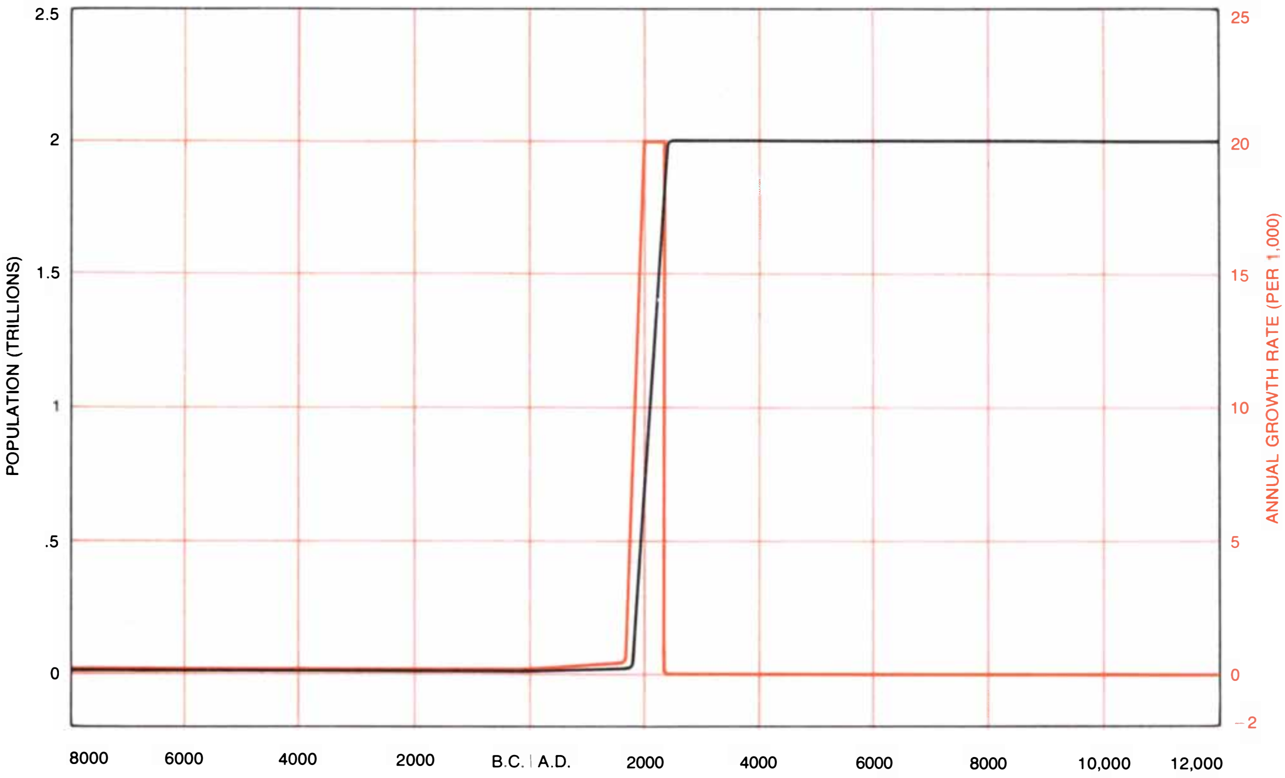
Whatever the cause of the initial decline in the death rate, there is no doubt that subsequent improvements in sanitation, public health and medicine made possible further reductions during the 19th century; indeed, the process continues today. It is equally clear that the reduction in mortality was dependent on the increased availability of food and other material resources. This rise in living standards was in turn brought about by the extension of cultivation, particularly in the Western Hemisphere, by increased productivity in both agriculture and industry and by the development of efficient trade and transportation.

The decline in the birth rate that eventually followed the decline in the death rate in the more developed countries was, with the exception of late-19th-century Ireland, almost entirely a decline in the fertility of married couples and can be attributed directly to the practice of contraception and abortion. The reduction in fertility was not a result of the invention of new contraceptive techniques, however. Among selected Americans married before 1910, English couples interviewed in the 1930's and couples surveyed in France and several eastern European nations after World War II, the principal method of birth control was *coitus interruptus,* a technique that had always been available. The birth rate declined because the perceived benefits and liabilities of having more children had changed, and perhaps also because the couples' view of the propriety of preventing births had been modified.

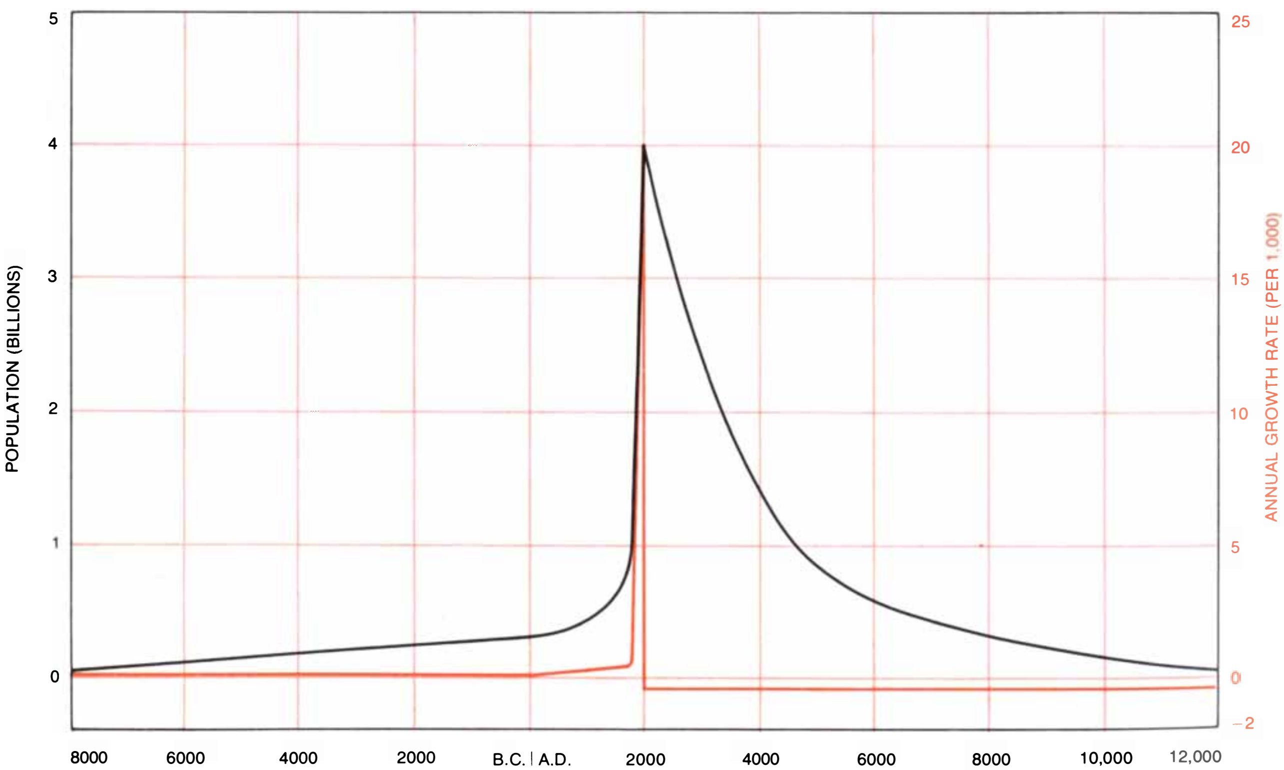
Reduced fertility can be considered one of the consequences of the charac-



THE DEMOGRAPHIC TRANSITION, represented schematically here, is the central event in the recent history of the human population. It begins with a decline in the death rate (*black*), precipitated by advances in medicine (particularly in public health), nutrition or both. Some years later the birth rate (*color*) also declines, primarily because of changes in the perceived value of having children. Before the transition the birth rate is constant but the death rate varies; afterward the death rate is constant but the birth rate fluctuates. The demographic transition usually accompanies the modernization of nations; it began in Europe and the U.S. late in the 18th century and early in the 19th, but in the underdeveloped nations it began only much later, often in the 20th century. In the developed countries the transition is now substantially complete, but in much of the rest of the world only mortality has been reduced; the fertility rate remains high. In the interim between the drop in mortality and fertility population has increased rapidly.



RETURN TO NEAR-ZERO GROWTH does not necessarily imply a low future population. If the rate of increase (*color*) were to remain at its present level until the year 2300, then immediately drop to zero, the average growth rate over the next 10,000 years would be as low as it has been during the past 10,000. Population, however (*black*), would reach two trillion, 500 times the present one.



IMMEDIATE REDUCTION of the growth rate to —.2 would result in a slow decline, eventually returning the world population to its level of 10,000 years ago: about eight million, or 1/500th of the present population. Under these circumstances too the rate of change in the size of the population during the next 10,000 years would differ from zero no more than it did during the past 10,000.

teristics by which the more developed countries are defined. In an urban, industrial society the family is no longer the main locus of economic activity, nor are children the expected means of support in old age. In an agrarian, preindustrial society, on the other hand, the family is a basic economic unit and sons are a form of social security. Moreover, in the less developed countries the costs of raising and educating a child are minimal; indeed, a child may contribute to the welfare of the family from an early age. In the industrial society child labor is prohibited, education is compulsory and it often extends through adolescence. These conditions conspire to discourage couples in the more developed countries from having large families, whereas in long-established agrarian societies social norms supporting childbearing tend to be perpetuated.

In the less developed countries the estimated rate of population growth was virtually zero until about 200 years ago, when a moderate rate of increase, about four per 1,000, was apparently induced by a reduction in mortality [*see illustration on page 47*]. The cause of this reduced death rate is uncertain. Durand has suggested that the interchange of staple foods between regions that had previously been isolated might have contributed to population growth in Asia and in Europe as well. In particular the introduction of the potato into Europe and of maize and the sweet potato into China have been cited as possible contributing factors.

Since the aggregate population of the less developed countries includes many large areas that have not been reliably enumerated, a description of the historical course of population growth in these countries is subject to much uncertainty. A slight reduction in the average rate of increase in the latter half of the 19th century, for example, can be attributed entirely to an estimated zero rate of growth in China, and that estimate is based on uncertain data. There is no doubt, however, that in the poorer nations rapid growth began in the 1920's, 1930's and 1940's, and that since World War II the population increase has accelerated dramatically.

The enormous recent growth of the populations of the less developed nations can be interpreted in terms of the demographic transition, but some parts of the process have been more rapid and more extreme than they were in the industrial nations; moreover, the transition is not yet complete, and its future course cannot be predicted. Mortality has dropped precipitously, but fertility has so far remained unchanged or declined only moderately. In the combined populations of the less developed countries the number of births per woman is about 5.5, and the average duration of life is more than 50 years, yielding an annual growth rate of about 25 per 1,000. Since World War II mortality in the less developed countries has fallen much more quickly than it did in 19th-century Europe, largely because modern technology, and particularly medical technology, can be imported more rapidly today than it could be discovered and developed 100 years ago. Insecticides, antibiotics and public health measures that were unknown during the European demographic transition are now commonplace in the less developed countries.

According to estimates prepared by the UN, the average duration of life in the less developed areas has risen from 32 to 50 years during the past three decades, an increase of 56 percent. During the same period the birth rate is estimated to have declined by no more than 7 to 8 percent. The actual fall in fertility is in fact even less, by about 4 percent, since demographic changes have reduced the proportion of women in the childbearing years. (Although the fertility of the less developed countries as a group remains very high, there are some countries where the birth rate has fallen significantly—by from 25 to 50 percent—and very rapidly. They include Hong Kong, Singapore, Taiwan, South Korea, West Malaysia, Barbados, Chile, Cuba, Jamaica, Trinidad and Tobago, Puerto Rico and Mauritius. According to reports from travelers, there has also been a decline in fertility in China, particularly in the cities.)

The present rapid growth in the world population is a result of a high rate of increase in the less developed areas and a moderate rate in the rest of the world. According to projections prepared by the UN, more than 90 percent of the increase in population to be anticipated by 2000 will be contributed by the less developed nations, even though a large reduction in fertility in these countries is expected in the next 25 years. The future course of the world's population depends largely on demographic trends in these countries.

The events of the demographic transition provide no sure way of calculating when or how quickly fertility will decline in the less developed nations. The experience of the industrial world is not a satisfactory basis for prediction. The history of the Western population during the past 200 years suggests that vital rates normally fall as a concomitant of modernization, but it provides no checklist of advances in literacy, mortality reduction and urbanization that would enable one to estimate when fertility will fall. In the more developed world there are instances of large reductions in fertility in populations that were still rural, mostly illiterate and still subject to moderately high mortality, as in the Garonne valley in southwestern France before 1850. In other instances fertility did not decline until after education was almost universal, the population was mostly urban and agriculture had become the occupation of a small minority, as in England and Wales.

The present rate of world population increase—20 per 1,000—is almost certainly without precedent, and it is hundreds of times greater than the rate that has been the norm for most of man's history. Without doubt this period of growth will be a transitory episode in the history of the population. If the present rate were to be maintained, the population would double approximately every 35 years, it would be multiplied by 1,000 every 350 years and by a million every 700 years. The consequences of sustained growth at this pace are clearly impossible: in less than 700 years there would be one person for every square foot on the surface of the earth; in less than 1,200 years the human population would outweigh the earth; in less than 6,000 years the mass of humanity would form a sphere expanding at the speed of light. Considering more realistic limits for the future, if the present population is not multiplied by a factor greater than 500 and thus does not exceed two trillion, and if it does not fall below the estimated population of preagricultural society, then the rate of increase or decrease during the next 10,000 years must fall as close to zero as it was during the past 10,000 years [*see illustrations on opposite page*].

Arithmetic makes a return to a growth rate near zero inevitable before many generations have passed. What is uncertain is not that the future rate of growth will be about zero but how large the future population will be and what combination of fertility and mortality will sustain it. The possibilities range from more than eight children per woman and a life that lasts an average of 15 years, to slightly more than two children per woman and a life span that surpasses 75 years.